Neutrino Scattering, Parity Breaking, and the Bounded Electron-Cloud Embedding

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Abstract

We formulate a coupling-matrix model in which the alternating sum over fermionic blocks is fixed by the quantum spin of the composite source. The muon and tau neutrino channels correspond to the two spin-graded sectors of that alternating sum. The electron neutrino is instead an electron-cloud embedding: a bounded elliptic projection $\Pi_e(t, r) \in [0, 1]$ on the torus $\mathbb{C}/(\pi\mathbb{Z} + e\mathbb{Z})$. In the unbroken parity sector the electron-neutrino cloud does not scatter with matter: it has no charge, does not ionize, and does not couple to the magnetic field. Scattering is opened only when the 720-cell parity split $6! = 360 + 360$ is broken. After parity breaking, off-diagonal couplings g_{ij} to composite matter become nonzero and the cloud participates in scattering. Even when it does not scatter, the coupling matrix makes the electron-neutrino cloud the underlying structure of the point-particle background.

1 Constants and Coupling Data

The fundamental finite-step scale is

$$\alpha = \frac{1}{\pi - e} \approx 2.3623. \tag{1}$$

The coupling matrix between two blocks i, j is

$$g_{ij} = w_{ij}\kappa_{ij}, \tag{2}$$

with chiral bond weight

$$w_{ij} = \frac{|\chi_i - \chi_j|}{\pi} \in [0, 1], \qquad \chi(Z) = \arg \Pi_6(Z), \tag{3}$$

and travel-length coupling

$$\kappa_{ij} = \frac{\sqrt{L_i L_j}}{L_{\text{ref}}} \in [0, 1], \qquad L_i = \int_{\gamma_i} \sqrt{\mathrm{d}t^2 + \mathrm{d}r^2}. \tag{4}$$

No normalization is applied to the readout; the coupling matrix itself is the observable.

2 Levi–Civita Embedding and the Spin-Graded Alternating Sum

For a 6×6 complex block Z, define the rank-6 Levi–Civita projection

$$\Pi_6(Z) = \sum_{\sigma \in S_6} \epsilon(\sigma) \prod_{k=1}^{6} Z_{k,\sigma(k)} = \det(Z). \tag{5}$$

The chirality of the block is

$$\chi(Z) = \arg \Pi_6(Z). \tag{6}$$

Let C be a composite particle with total spin J_C. Define the spin grade

$$\sigma_\ell(J_C) = (-1)^{2J_C+\delta_\ell}, \qquad \ell = \mu, \tau, \tag{7}$$

where the two values $\delta_\mu = 0$ and $\delta_\tau = 1$ select the muon and tau neutrino channels. The alternating sum over the block family is then fixed by the spin of the composite source:

$$S_{\nu_\ell} = \sum_{n=1}^{N} (-1)^{n+1} \sigma_\ell(J_C) \big|\Pi_6(Z_n)\big|, \qquad \ell = \mu, \tau. \tag{8}$$

Thus the muon and tau neutrino sectors are the two spin-graded alternating-sum embeddings of the same Levi–Civita block family.

The corresponding invariant mass is

$$m_{\nu_\ell} = |S_{\nu_\ell}| \exp(-H_{\nu_\ell}), \qquad H_{\nu_\ell} = -\alpha \frac{|S_{\nu_\ell}|}{N} \log\left(\frac{|S_{\nu_\ell}|}{N}\right). \tag{9}$$

3 Electron Neutrino as a Bounded Elliptic Cloud

The electron neutrino is not represented by the alternating sum above. It is represented by the electron-cloud embedding of Figure 3.5,

$$\Pi_e(t, r) = \frac{1}{2}\left[1 + \cos\left(\frac{2\pi t}{\pi}\right)\cos\left(\frac{2\pi r}{e}\right)\right] = \frac{1}{2}\left[1 + \cos(2t)\cos\left(\frac{2\pi r}{e}\right)\right], \tag{10}$$

on the torus

$$\mathbb{C}/(\pi\mathbb{Z} + e\mathbb{Z}). \tag{11}$$

By construction,

$$0 \le \Pi_e(t, r) \le 1. \tag{12}$$

The elliptic curve is kept bounded between 0 and 1; it is the electron-cloud embedding, not a free propagating charged field.

4 Parity Protection and Non-Scattering

The 720-cell has the parity split

$$6! = |A_6| + |\text{odd complement}| = 360 + 360. \tag{13}$$

Define the parity operator on the block family by

$$PZ_n = \begin{cases} +Z_n, & \sigma_n \in A_6, \\ -Z_n, & \sigma_n \notin A_6. \end{cases} \tag{14}$$

In the unbroken parity sector the admissible coupling is restricted by the even-even mask

$$M_{ij} = \begin{cases} 1, & i \equiv j \equiv 0 \pmod 2, \\ 0, & \text{otherwise.} \end{cases} \tag{15}$$

Thus

$$G_{ij}^{\text{even}} = M_{ij} g_{ij}, \tag{16}$$

and all odd-index pairs are identically zero.

In this sector the electron-neutrino cloud does not scatter with matter. Since it carries no charge in the embedding,

$$q_{\nu_e} = 0, \tag{17}$$

the electromagnetic Lorentz force vanishes:

$$\mathbf{F}_{\text{EM}} = q_{\nu_e} \left(\mathbf{E} + \mathbf{v} \times \mathbf{B} \right) = 0. \tag{18}$$

Therefore the electron-neutrino cloud:

- does not ionize matter;
- does not scatter with the magnetic field;
- does not couple to electromagnetism;
- remains a non-scattering background in the unbroken parity sector.

5 Parity Breaking and Scattering with Composite Matter

Parity breaking occurs when the A_6 sector mixes with the odd complement. Then the even-even mask no longer annihilates the odd entries, and the off-diagonal coupling

$$g_{ij} = w_{ij} \kappa_{ij} \tag{19}$$

becomes nonzero for odd i or odd j. The reduced flow acquires the coupling force

$$\dot{Z}_i = \alpha \Im\left(\frac{\mathrm{d}\zeta}{\mathrm{d}t} \right) Z_i + \sum_j g_{ij} \left(Z_j - Z_i \right). \tag{20}$$

Scattering with composite matter is then controlled by the coupling strength,

$$\sigma_{\text{scat}} \ \propto \ |g_{ij}|^2 = w_{ij}^2 \kappa_{ij}^2. \tag{21}$$

Only after parity breaking does the electron-neutrino cloud "interact" or scatter with composite matter. Otherwise it remains electromagnetically inert.

6 Underlying Point-Particle Cloud

Even in the non-scattering sector, the coupling matrix makes the electron-neutrino cloud the underlying structure of the point-particle background. The travel lengths

$$L_i = \int_{\gamma_i} \sqrt{\mathrm{d}t^2 + \mathrm{d}r^2} \tag{22}$$

and the coupling kernel

$$\kappa_{ij} = \frac{\sqrt{L_i L_j}}{L_{\text{ref}}} \tag{23}$$

define the point-particle cloud. The transcendental search flow is

$$\zeta_\mu(s) = \sum_{k\geq 1} \mu(k) k^{-s}, \qquad \frac{\mathrm{d}s}{\mathrm{d}t} = -\zeta'_\mu(s), \tag{24}$$

with readout performed through the coupling matrices. Thus the point-particle cloud is not a test charge moving in an external field; it is the singularity structure of the coupling matrix itself.

7 Summary

1. The alternating sum over fermionic blocks is spin-graded by the composite source. The muon and tau neutrino channels correspond to the two spin-graded sectors.

2. The electron neutrino is an electron-cloud embedding:

   $$\Pi_e(t, r) \in [0, 1]$$

   on $\mathbb{C}/(\pi\mathbb{Z} + e\mathbb{Z})$, the bounded elliptic curve of Figure 3.5.

3. In the unbroken parity sector the electron-neutrino cloud does not scatter with matter: no ionization, no magnetic scattering, no electromagnetic coupling.

4. Parity breaking in the 720-cell,

   $$720 = 360 + 360,$$

   opens the odd sector. Off-diagonal couplings g_{ij} become nonzero, and scattering with composite matter is enabled.

5. Due to the coupling matrix $g_{ij} = w_{ij}\kappa_{ij}$, the electron-neutrino cloud is the underlying structure of the point-particle cloud even when it is non-scattering.